

JEE MAIN REVISION MASTER: TRIGONOMETRIC EQUATIONS OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 59 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Trigonometric Functions And Equations - Core Application	59 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Functions And Equations - Core Application (Very High Yield — 59 PYQs)

Core Principle: Key problem pattern in Trigonometric Functions And Equations.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$. Then $\inf_{\theta \in S} \theta$ is equal to :..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "The sum of all the integral values of p such that the equation $3\sin^2x + 12\cos x - 3 = p$, $x \in [0, \pi]$, has at least one solution, is:..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "Let $S = \{x \in [-\pi, \pi] : \sin x (\sin x + \cos x) = a, a \in \mathbb{R}\}$. Then $n(S)$ is equal to:..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$. Then $\sum \theta \in S$ is equal to :

- (A) $-(2\pi / 3)$
- (B) $-(4\pi / 3)$
- (C) $(2\pi / 3)$
- (D) $(4\pi / 3)$

Correct Answer: None

Solution Analysis: Given, $\cos\theta + 1 = \sqrt{3}\sin\theta$ We have to find all $\theta \in (-2\pi, 2\pi)$ satisfying this equation, and then calculate their sum. Rewrite the equation as $1 + \cos\theta = \sqrt{3}\sin\theta$ Now use the standard identities: $1 + \cos\theta = 2\cos^2(\theta / 2)$ and $\sin\theta = 2\sin(\theta / 2)\cos(\theta / 2)$ Substituting these, $2\cos^2(\theta / 2) = \sqrt{3} \cdot 2\sin(\theta / 2)\cos(\theta / 2)$ So, $2\cos(\theta / 2)(\cos(\theta / 2) - \sqrt{3}\sin(\theta / 2)) = 0$ Hence, either Case 1: $\cos(\theta / 2) = 0$ So, $(\theta / 2) = (\pi / 2) + n\pi$ which gives